

Hyuksoon Chang

📍 Santa Cruz, California ✉ hyuksoon@ucsc.edu

Education

Ph.D. University of California, Santa Cruz	Sep 2026 –
Computer Science and Engineering (advised by Mohsen Lesani)	
M.S. Pohang University of Science and Technology	Aug 2023 – Feb 2026
Computer Science and Engineering (advised by Kyungmin Bae)	
B.S. Chungbuk National University	Mar 2019 – Mar 2023
Computer Engineering (advised by Kyuri Jo)	

Publications

Model Checking of WebAssembly Software Using Maude	2025
Hyuksoon Chang, Kyungmin Bae. Korea Computer Congress(2025) 🔗	
Robustness Analysis of Convolutional Neural Networks for Circle Image Classification by Explainable Artificial Intelligence	2021
Jungho Park*, Hyuksoon Chang*, Kyuri Jo. Korea Computer Congress(2021) 🔗	
Inferring transcriptomic cell states and transitions from time series transcriptome data	2021
Jo K, Sung IY, Lee D, Jang H, Kim S. Scientific Reports(2021) 🔗	

Projects & Experience

Model Checking of WebAssembly Software with WASM-DSL	2024 - 2026
- Developing a model-checking framework for WebAssembly software to verify distributed-system properties - Implemented the official WebAssembly specification in the Maude rewriting-logic framework to execute Wasm modules and perform LTL-based model checking. - Designed compilation rules that translate the WASM-DSL language into Maude rewrite theories, ensuring automatic conformance with the WebAssembly specification.	
BML Lab: Relation prediction of Disease and Transcriptome using LRP method	2020 - 2021
- Participated in the Bioinformatics and Machine Learning Lab as an undergraduate researcher and founding member - Developed a disease-transcriptome prediction model using PyTorch, integrating Layer-wise Relevance Propagation (LRP) to interpret model decisions. Optimized computational efficiency for experimental data processing	
MimicStudio: Web service that automatically generates 3D product models	2022
- Led the team and implemented the neural-rendering server with Instant-NGP - Implemented an upscaling algorithm to improve the resolution of 3D videos	
Cirkadian: Healthcare devices based on circadian rhythm	2022.04 - 2022.07
- Developed a cross-platform mobile application using Flutter to interface with circadian illuminometers (light sensors) and control circadian lamps remotely - Improved application stability and user experience (UI/UX) by resolving bugs and implementing iterative design improvements	

COSMIC: AI-Driven Web Projects Club

2020 - 2022

- Founded and led the club; built web projects including an AI-based security-camera service and an undergraduate course-recommendation platform
- Designed and supervised several instructional toy projects to teach web frameworks such as React, FastAPI and Django

TEACHING

CSED415: Computer Security (TA) [🔗](#)

2025 Spring

5110160: Machine Learning (TA)

2021 Fall

0622014: Applied Computer Programming (TA)

2020 Fall, 2021 Fall

0914002: Fundamental Computer Programming (TA)

2021 Spring, 2022 Spring

AWARDS & FUNDING

Master's Research Fellowship — National Research Foundation of Korea

2024–2025

Lotte Scholarship

2020–2023

Daeshin Freight Scholarship

2020–2023

Software Maestro Program Grant, 13th Cohort — Ministry of Science and ICT

2022

Excellence Scholarship — Software-Centered University Program, CBNU

2020–2022